
SINJINI BANERJEE

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TECHNICAL SKILLS

Software tools	Matlab/Simulink Jupyter Notebook Eclipse Colab Tensorflow Pytorch Jax Atom/ Uber-Juno Gurobi Optimiser Android Studio Onshape
Programming Languages	Julia Python C SQL
Hardware Tools	Arduino Programmable Logic Controllers (PLC)

EDUCATION

Rutgers University, NJ	Ph.D. – Electrical Engineering (Specialization: Information & Signal Processing)	GPA 3.93/4	Sept 2020 – Present
University at Buffalo – SUNY	M.S. - Electrical Engineering, Thesis: Signal optimization	GPA 3.62/4	June 2019
Heritage Institute of Technology, Kolkata, India	B.Tech - Applied Electronics & Instrumentation Engineering	GPA 8.35/10	July 2016

WORK EXPERIENCE

Graduate Research Assistant, Department of Electrical and Computer Engineering, Rutgers University Sept 2020 – Present

- **Measuring training variability from stochastic optimization using robust nonparametric testing**, (Utilized TensorFlow, Pytorch, Jax)
 - Developed a robust hypothesis testing framework to quantify model variability due to stochastic optimization in deep learning models, in collaboration with Pacific Northwest National Laboratory.
 - Introduced a novel model selection metric that goes beyond traditional accuracy measures to better capture model similarity and to identify the number of models to ensemble for reliable predictions.
 - Built, optimized, and fine-tuned over 1000 deep learning models (Feedforward Neural Networks, MLPs, CNNs, Vision Transformer, BERT) using high-performance computing (HPC) infrastructure provided by Rutgers' Office of Advanced Research Computing.
- **Mitigating risks associated with prediction inconsistency of equally accurate deep net models in machine learning model markets**, (Utilized TensorFlow, Pytorch, Jax)
 - Designed and developed a testing tool to identify clients negatively impacted by prediction inconsistency with 95% accuracy in high stakes application domains like credit-lending.
 - Minimized costs associated with providing human intervention by establishing high probabilistic guarantee on true positives.
 - Developed an algorithm to generate robust counterfactuals for clients negatively impacted by prediction inconsistency.

Graduate Teaching Assistant, Department of Electrical and Computer Engineering, Rutgers University Sept 2024– Present

- Taught and supported over 90 first-year engineering students in developing core technical and analytical skills, including 3D CAD modeling (Onshape), MATLAB programming, and data analysis with Excel.
- Developed hands-on coding assignments and real-world engineering challenges that strengthened students' understanding of control flow, functions, vectorized operations, and data plotting in MATLAB.
- Guided students through debugging sessions and best practices for writing efficient, readable code, laying a strong foundation for future work in data science, machine learning, and computational modeling.

INTERNSHIP

Intern, Department of Electrical Engineering, University at Buffalo, (Utilized Julia)

Aug 2019 – May 2020

- Developed a novel outlier-robust Recursive Least Squares (RLS) algorithm by integrating sparsity-aware modeling of outliers with a hierarchical optimization framework (HO-RLS), enhancing robustness in noisy environments.
- Benchmarked performance against state-of-the-art robust RLS variants, showing superior estimation accuracy and faster adaptation in both stationary and non-stationary signal processing scenarios.
- Validated algorithm effectiveness and improved performance through extensive synthetic experiments and the implementation of parallel computing on clusters available at Center for Computation Research at University at Buffalo.

PAPERS

- Banerjee, S., Marrinan, T., Cannon, R., Chiang, T., & Sarwate, A. D. (2024). Measuring training variability from stochastic optimization using robust nonparametric testing. arXiv preprint arXiv:2406.08307 (Currently under review in IEEE Journal of Selected Topics in Signal Processing). Sept 2023
- Slavakis, K., & Banerjee, S. (2019). Robust hierarchical-optimization RLS against sparse outliers. IEEE Signal Processing Letters, 27, 171-175. Jan 2020

ACADEMIC PROJECTS

Understanding tensor decomposition for spectral unmixing in hyperspectral images, (Utilized Matlab)

Sept 2021 – Sept 2022

- Applied spectral analysis techniques to process and analyze high-dimensional hyperspectral image data.
- Implemented efficient spectral unmixing through low-rank tensor decomposition to extract endmembers and generate abundance maps, with applications in remote sensing and environmental monitoring.

Musical Instrument Recognition using harmonics

Oct 2017 – Dec 2017

- Used cepstral analysis to identify, study, and characterize individual notes of two different musical instruments, flute and piano, in the reverse frequency domain.

Classification of cancer subgroups using microarray gene expression data

Sept 2016 – Mar 2017

- Used particle swarm optimization and adaptive K-nearest neighborhood technique on lung cancer data to classify cancer subgroups.
- Utilized t-test method for dimensionality reduction.
- Identified 14 genes that can be efficiently exploited for high accuracy diagnostic prediction.

WORKSHOPS

- Bellairs Workshop on Machine Learning and Statistical Signal Processing for Data on Graphs. Jan 2023
- DIMACS Workshop on Modeling Randomness in Neural Network Training: Mathematical, Statistical, and Numerical Guarantees. June 2024